

ABIGAIL PAULUS

Ambitious, detail-oriented engineering student focused on prototyping, aspiring to be a product designer

954-798-7482
abbypaulus@icloud.com
abbypaulus.github.io/links

EDUCATION

Bachelor of Science in Engineering Technology | Daytona State College, Daytona, FL

Aug 2026 – Present

Associates of Arts | Florida Atlantic University, Boca Raton, FL

June 2024 – Aug 2026

Awards: \$1,000 CSYS Scholarship (2024), Outstanding Engineering Achievement Award (Fall 2024, 2023, 2022), Cambridge AICE Diploma (2023)

SKILLS

Certifications: Solidworks CSWA, DraftSight Associate, AutoCAD, Inventor, Fusion360, Microsoft Office Specialist, Adobe Photoshop

Manufacturing: FDM 3D Printing, Soldering, Calipers, Power/Hand Tools, Drill Press, Woodworking, CNC Programming, Weldment, GD&T

Software: SketchUp, RapidHarness, Siemens NX, Solidworks PDM, AS400, ERP, Illustrator, Inkscape, Cura Slicer, Altium Designer, Blender

EXPERIENCE

Product Design Engineer, Shark Beauty NPD Intern | SharkNinja, Needham, MA

May 2026 – Present

- Build and test 14 quick turnaround, low fidelity prototypes, weighed and working to test different forms and basic functionality
- Collaborate cross-functionally with R&D and PD to define solutions that meet functional requirements while enhancing builds
- Conduct 3 internal user tests analyzing prototypes and baselining against competitors, iterating goals for future prototype builds
- Assemble heat pattern test rigs weekly to analyze prototypes using thermocouples and a Data Acquisition (DAQ) device
- Post-process data from CSV data and FLIR cameras to interpret thermal mapping and gather delta of different configurations
- Modify and tear-down competitor units to reduce touch temps of prototype cap meeting threshold of 60c from unmodified 93c

Programming Teacher (Contract) | Code/Art, Davie, FL

Feb 2024 – Present

- Instructing 10+ students within 1–2-hour lesson using programs such as: Scratch, p5.js, Tinkercad, JavaScript, HTML/CSS
- Verifying students' comprehension of lessons, conveying directions effectively to educators, and ensuring a supportive space

Supply Chain Documentation Intern | Dometic, Pompano Beach, FL

Oct 2025 – May 2026

- Perform duty drawback to recover ~\$500k in tariff reimbursement, developing and maintaining SOP for manufacturing audit trail
- Organize 13,200 finished good shipment trade documentation (import and export), using pivot tables within Excel for navigation

CAD Drafter/Designer Intern | Dometic, Pompano Beach, FL

May 2025 – Sept 2025

- Drafted, maintained, and updated 75+ Draftsight, RapidHarness and Solidworks drawings for components, layouts, assemblies
- Communicated with test engineers to produce custom crate drawings using weldments in Solidworks to manufacture in house

Mechanical Engineer Intern | Q'Straint, Oakland Park, FL

June 2024 – Aug 2024

- Designed metal support brackets for in-house production using SolidWorks' sheet metal tool and design intent
- Updated 10+ drawings in Solidworks, fulfilled 2 work orders, and tracked 200+ unused inventory parts using Excel

CAD Engineer Intern | Q'Straint, Oakland Park, FL

June 2023 – July 2023

- Revised 10+ outdated technical drawings to match current standards, replacing to use existing inventory, reducing cost
- Authored 6 pages of instruction manual using Word, creating template for similar components across the product line

PROJECTS

Electric Campervan Conversion *Personal Project*

Oct 2025 – Present

Developing modular plumbing, electrical, and HVAC systems for electric Chevy Brightdrop 600 van chassis to convert into campervan, designed using Siemens NX, with 15kW of additional onboard power, 35 gallons of water, charged by 600W of solar

Digital Audio Player *Personal Project*

July 2024 – Present

MP3 player reverse engineered using Raspberry Pi with custom PCBA and CNC machined chassis practicing DFM/DFA (design for manufacturing/assembly), research consumer needs, iterating/refining 2 different prototypes while prioritizing most valued features

Blind Spot Sensor for Motorcycle/Bicycle *Capstone Project*

Dec 2023 – Feb 2024

Designed concept in Fusion360, with 4 FDM prototypes, ensuring proper fit of existing components found in premade starter kit to develop Arduino based modular system that detect objects within a biker's blind-spot, with vibrating motor module on the helmet

Bar-top Display Stand for Teacher's Lounge *Club Project*

Mar 2023 – Oct 2023

Arranged cut list using Excel and drafted assembly drawings within SketchUp, while managing team of 3 with effective communication, to design and build custom wooden bar-top, optimizing material selection, leaving 30% of \$1000 budget

LEADERSHIP

Secretary Committee Member, Society of Women Engineers (Spring '26) | **Technical Outreach and Community Help Chair**, National Society of Black Engineers ('25-26) | **Academic, Professional & Athletic Chair**, Alpha Omega Epsilon ('25-26) | **Volunteer Soccer Coach**, Coral Springs Youth Soccer ('22-25) | **Peer Mentor** Society of Hispanic Professional Engineers (Fall '25) | **Internal Development Chair**, American Society of Mechanical Engineers ('25) | **President**, Southeastern Consortium for Minorities in Engineering ('22-24)